

Polynomial Hierarchy

“A polynomial-bounded version of Kleene’s Arithmetic Hierarchy becomes trivial if $\mathbf{P} = \mathbf{NP}$.”

Karp, 1972



Larry Stockmeyer and Albert Meyer introduced polynomial hierarchy.

1. Larry Stockmeyer and Albert Meyer. The Equivalence Problem for Regular Expressions with Squaring Requires Exponential Space. SWAT'72.

Synopsis

1. Meyer-Stockmeyer's Polynomial Hierarchy
2. Stockmeyer-Wrathall Characterization
3. Chandra-Kozen-Stockmeyer Theorem
4. Infinite Hierarchy Conjecture
5. Time-Space Trade-Off

Meyer-Stockmeyer's Polynomial Hierarchy

Problem Beyond NP

Meyer and Stockmeyer observed that MINIMAL does not seem to have short witnesses.

$$\text{MINIMAL} = \{\varphi \mid \varphi \text{ DNF} \wedge \forall \text{ DNF } \psi. |\psi| < |\varphi| \Rightarrow \exists u. \neg(\psi(u) \Leftrightarrow \varphi(u))\}.$$

Notice that $\overline{\text{MINIMAL}}$ can be solved by an NDTM that queries SAT a polynomial time.

- Why DNF?

$$\begin{aligned}\mathbf{P}^{\mathcal{C}} &= \bigcup_{A \in \mathcal{C}} \mathbf{P}^A, \\ \mathbf{NP}^{\mathcal{C}} &= \bigcup_{A \in \mathcal{C}} \mathbf{NP}^A.\end{aligned}$$

Meyer-Stockmeyer's Definition

The complexity classes $\Sigma_i^P, \Pi_i^P, \Delta_i^P$ are defined as follows:

$$\begin{aligned}\Sigma_0^P &= \mathbf{P}, \\ \Sigma_{i+1}^P &= \mathbf{NP}^{\Sigma_i^P}, \\ \Delta_{i+1}^P &= \mathbf{P}^{\Sigma_i^P}, \\ \Pi_i^P &= \overline{\Sigma_i^P}.\end{aligned}$$

The following hold:

- ▶ $\Sigma_i^P \subseteq \Delta_{i+1}^P \subseteq \Sigma_{i+1}^P$,
- ▶ $\Pi_i^P \subseteq \Delta_{i+1}^P \subseteq \Pi_{i+1}^P$.

Notice that $\Pi_{i+1}^P = \mathbf{coNP}^{\Sigma_i^P}$ by definition.

The **polynomial hierarchy** is the complexity class **PH** = $\bigcup_{i \geq 0} \Sigma_i^P$.

Natural Problem in the Second Level

“Synthesizing circuits is exceedingly difficult. It is even more difficult to show that a circuit found in this way is the most economical one to realize a function. The difficulty springs from the large number of essentially different networks available.”

Claude Shannon, 1949

Umans showed in 1998 that the following language is Σ_2^P -complete.

$$\text{MIN-EQ-DNF} = \{ \langle \varphi, k \rangle \mid \varphi \text{ DNF} \wedge \exists \text{ DNF } \psi. |\psi| \leq k \wedge \forall u. \psi(u) \Leftrightarrow \varphi(u) \}.$$

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- ▶ $\overline{\text{MIN-EQ-DNF}}$ is the problem referred to by Shannon.
 - ▶ The complexity of MINIMAL, as well as $\overline{\text{MINIMAL}}$, is not known.

Natural Problem in the Second Level

SUCCINCT SET COVER:

Given a set $S = \{\varphi_1, \dots, \varphi_m\}$ of 3-DNF's and an integer k , is there a subset $S' \subseteq \{1, \dots, m\}$ of size at most k such that $\bigvee_{i \in S'} \varphi_i$ is a tautology?

This is another Σ_2^P -complete problem.

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1. C. Umans. The Minimum Equivalent DNF Problem and Shortest Implicants. JCSS, 597-611, 2001. Preliminary version in FOCS 1998.

Natural Problem in the Second Level

EXACT INDSET refers to the following problem:

$\{\langle G, k \rangle \mid \text{the largest independent sets of } G \text{ are of size } k\}$.

It is in Δ_2^P and is **DP**-complete.

$L \in \mathbf{DP}$ if $L = L_0 \cap L_1$ for some $L_0 \in \mathbf{NP}$ and some $L_1 \in \mathbf{coNP}$. Clearly

NP, coNP $\subseteq$ **DP**.

Stockmeyer-Wrathall Characterization

In 1976, Stockmeyer defined Polynomial Hierarchy in terms of alternation of quantifier and Wrathall proved that it is equivalent to the original definition.

1. Larry Stockmeyer. The Polynomial-Time Hierarchy. Theoretical Computer Science, 3:1-22, 1976.
2. Celia Wrathall. Complete Sets and the Polynomial-Time Hierarchy. Theoretical Computer Science. 3:23-33, 1976.

Logical Characterization

The following result generalizes the logical characterization of NP problems.

Theorem. Suppose $i \geq 1$.

- $L \in \Sigma_i^P$ iff there exists a P-time TM $\mathbb{M}$ and a polynomial q such that for all $x \in \{0, 1\}^*$,

$$x \in L \text{ iff } \exists u_1 \in \{0, 1\}^{q(|x|)} \forall u_2 \in \{0, 1\}^{q(|x|)} \dots Q_i u_i \in \{0, 1\}^{q(|x|)}. \mathbb{M}(x, \tilde{u}) = 1.$$

- $L \in \Pi_i^P$ iff there exists a P-time TM $\mathbb{M}$ and a polynomial q such that for all $x \in \{0, 1\}^*$,

$$x \in L \text{ iff } \forall u_1 \in \{0, 1\}^{q(|x|)} \exists u_2 \in \{0, 1\}^{q(|x|)} \dots Q_i u_i \in \{0, 1\}^{q(|x|)}. \mathbb{M}(x, \tilde{u}) = 1.$$

1. Celia Wrathall. Complete Sets and the Polynomial-Time Hierarchy. Theoretical Computer Science. 3:23-33, 1976.

Proof of Wrathall Theorem

Let $\mathbb{M}$ be a P-time TM and q a polynomial such that $x \in L$ if and only if

$$\exists u_1 \in \{0, 1\}^{q(|x|)} \dots Qu_{i+1} \in \{0, 1\}^{q(|x|)}. \mathbb{M}(x, u_1, \dots, u_{i+1}) = 1.$$

Given x an NDTM guesses a u_1 and asks if the following is true

$$\forall u_2 \in \{0, 1\}^{q(|x|)} \dots Qu_{i+1} \in \{0, 1\}^{q(|x|)}. \mathbb{M}(x, u_1, \dots, u_{i+1}) = 1.$$

By induction hypothesis the above formula can be evaluated by querying a Σ_i^P oracle.

Proof of Wrathall Theorem

Let L be decided by a P-time NDTM $\mathbb{N}$ with access to some oracle $A \in \Sigma_i^P$. Now by Cook-Levin Theorem, $x \in L$ if and only if

$\exists \tilde{z}. \exists c_1, \dots, c_m, a_1, \dots, a_k. \exists u_1, \dots, u_k. (\mathbb{N} \text{ accepts } x \text{ using choices } c_1, \dots, c_m$
and answers $a_1, \dots, a_k$ to the queries $u_1, \dots, u_k) \wedge (\bigwedge_{i \in [k]} a_i = 1 \Rightarrow u_i \in A)$
 $\wedge (\bigwedge_{i \in [k]} a_i = 0 \Rightarrow u_i \in \overline{A}),$

where $\tilde{z}$ are introduced by the Cook-Levin reduction. We are done by induction.

$\Sigma_i\text{SAT}$

Let $\Sigma_i\text{SAT}$ be the subset of TQBF that consists of all tautologies of the following form

$$\exists u_1 \forall u_2 \dots Q_i u_i. \varphi(u_1, \dots, u_i),$$

where $\varphi(u_1, \dots, u_i)$ is a **propositional** formula.

Theorem (Meyer and Stockmeyer, 1972). $\Sigma_i\text{SAT}$ is Σ_i^P -complete.

Proof.

Clearly $\Sigma_i\text{SAT} \in \Sigma_i^P$. The completeness is defined with regards to Karp reduction. $\square$

Theorem (Stockmeyer, Wrathall, 1976). $\mathbf{PH} \subseteq \mathbf{PSPACE}$.

Chandra-Kozen-Stockmeyer Theorem



Ashok Chandra, Dexter Kozen and Larry Stockmeyer introduced Alternating Turing Machines that give alternative characterization of complexity classes.

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1. Alternation. *Journal of the ACM*, 28(1):114-133, 1981.

Alternating Turing Machine

An **Alternating Turing Machine** (ATM) is an **NDTM** in which every state is labeled by an element of $\{\exists, \forall, \text{accept}, \text{halt}\}$.

We say that an ATM $\mathbb{A}$ **accepts** x if there is a subtree Tr of the execution tree of $\mathbb{A}(x)$ satisfying the following:

- ▶ The initial configuration is in Tr .
 - ▶ All leaves of Tr are labeled by **accept**.
 - ▶ If a node labeled by $\forall$ is in Tr , both children are in Tr .
 - ▶ If a node labeled by $\exists$ is in Tr , one of its children is in Tr .
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NDTM's are ATM's.

Complexity via ATM

For every $T : \mathbf{N} \rightarrow \mathbf{N}$, we say that an ATM $\mathbb{A}$ runs in $T(n)$ -time if for every input $x \in \{0, 1\}^*$ and for all nondeterministic choices, $\mathbb{A}$ halts after at most $T(|x|)$ steps.

- ▶ **ATIME**($T(n)$) contains L if there is a $cT(n)$ -time ATM $\mathbb{A}$ for some constant c such that, for all $x \in \{0, 1\}^*$, $x \in L$ if and only if $\mathbb{A}(x) = 1$.
- ▶ **ASPACE**($S(n)$) is defined analogously.

Example of ATM

TQBF is solvable by an ATM in quadratic time and linear space.

Complexity Class via ATM

$$\mathbf{AL} = \mathbf{ASPACE}(\log n),$$

$$\mathbf{AP} = \bigcup_{c>0} \mathbf{ATIME}(n^c),$$

$$\mathbf{APSPACE} = \bigcup_{c>0} \mathbf{ASPACE}(n^c),$$

$$\mathbf{AEXP} = \bigcup_{c>0} \mathbf{ATIME}(2^{n^c}),$$

$$\mathbf{AEXPSPACE} = \bigcup_{c>0} \mathbf{ASPACE}(2^{n^c}).$$

Theorem. Assume that the relevant time/space functions are constructible.

1. $\mathbf{NSPACE}(S(n)) \subseteq \mathbf{ATIME}(S^2(n))$.
 2. $\mathbf{ATIME}(T(n)) \subseteq \mathbf{SPACE}(T(n))$.
 3. $\mathbf{ASPACE}(S(n)) \subseteq \bigcup_{c>0} \mathbf{TIME}(c^{S(n)})$.
 4. $\mathbf{TIME}(T(n)) \subseteq \mathbf{ASPACE}(\log T(n))$.
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1. Savitch's proof. Recursive calls are implemented using $\forall$ -state. We need to assume that $S(n)$ is constructible in $S(n)^2$ time.
2. Traversal of configuration tree. Counters of length $T(n)$. We need to assume that $T(n)$ is also space constructible.
3. Depth first traversal of configuration graph.
4. Backward guessing ($\exists$) and parallel checking ($\forall$) in the configuration circuit.

Chandra-Kozen-Stockmeyer Theorem

$$\begin{array}{ccccccc} \mathbf{AL} & \subseteq & \mathbf{AP} & \subseteq & \mathbf{APSPACE} & \subseteq & \mathbf{AEXP} \dots \\ = & & = & & = & & = \dots \\ \mathbf{L} & \subseteq & \mathbf{P} & \subseteq & \mathbf{PSPACE} & \subseteq & \mathbf{EXP} \subseteq \mathbf{EXPSPACE} \dots \end{array}$$

Bounded Alternation

$L \in \Sigma_i \text{TIME}(T(n)) / \Pi_i \text{TIME}(T(n))$ if

L is accepted by an $O(T(n))$ -time ATM $\mathbb{A}$ with q_{start} labeled by $\exists/\forall$, and on every path the machine $\mathbb{A}$ may alternate at most $i - 1$ times.

Polynomial Hierarchy Defined via ATM

Theorem. For every $i \geq 1$, the following hold:

$$\Sigma_i^P = \bigcup_{c \geq 0} \Sigma_i \mathbf{TIME}(n^c),$$

$$\Pi_i^P = \bigcup_{c \geq 0} \Pi_i \mathbf{TIME}(n^c).$$

Use the logical characterization.

Infinite Hierarchy Conjecture

Theorem. If $\mathbf{NP} = \mathbf{P}$ then $\mathbf{PH} = \mathbf{P}$.

Suppose $\Sigma_i^{\mathbf{P}} = \mathbf{P}$. Then $\Sigma_{i+1}^{\mathbf{P}} = \mathbf{NP}^{\Sigma_i^{\mathbf{P}}} = \mathbf{NP}^{\mathbf{P}} = \mathbf{NP} = \mathbf{P}$.

Theorem (Meyer and Stockmeyer, 1972). For every $i \geq 1$, if $\Sigma_i^P = \Pi_i^P$ then $\mathbf{PH} = \Sigma_i^P$.

Suppose $\Sigma_k^P = \Pi_k^P$. Then $\Sigma_{k+1}^P = \Sigma_k^P = \Pi_k^P = \Pi_{k+1}^P$.

Theorem. If there exists a language L that is **PH**-complete with regards to Karp reduction, then some i exists such that $\mathbf{PH} = \Sigma_i^P$.

If such a language L exists, then $L \in \Sigma_i^P$ for some i . Consequently every language in **PH** is Karp reducible to L .

Theorem. If $\mathbf{PH} = \mathbf{PSPACE}$, then $\mathbf{PH}$ collapses.

If $\mathbf{PH} = \mathbf{PSPACE}$, then TQBF would be $\mathbf{PH}$ -complete.

Infinite Hierarchy Conjecture. Polynomial Hierarchy does not collapse.

Many results in complexity theory take the following form

“If something is not true, then the polynomial hierarchy collapses”.

Time-Space Trade-Off

To summarize our current understanding of NP-completeness from an algorithmic point of view, it suffices to say that at the moment we **cannot** prove **either** of the following statements:

$\text{SAT} \notin \mathbf{TIME}(n),$

$\text{SAT} \notin \mathbf{SPACE}(\log n).$

We can however prove that SAT cannot be solved by any TM that runs in both linear time and logspace. Notationally,

$$\text{SAT} \notin \mathbf{TISP}(n, \log n).$$

Suppose $S, T : \mathbf{N} \rightarrow \mathbf{N}$. A problem is in

$$\mathbf{TISP}(T(n), S(n))$$

if it is decided by a TM that on every input x takes at most $O(T(|x|))$ time and uses at most $O(S(|x|))$ space.

Time-Space Tradeoff for SAT

Theorem. $\text{SAT} \notin \mathbf{TISP}(n^{1.1}, n^{0.1})$.

We show that $\mathbf{NTIME}(n) \not\subseteq \mathbf{TISP}(n^{1.2}, n^{0.2})$, which implies the theorem for the following reason:

1. Using Cook-Levin reduction a problem $L \in \mathbf{NTIME}(n)$ is reduced to a formula, every bit of the formula can be computed in **logarithmic** space and **polylogarithmic** time.
2. If $\text{SAT} \in \mathbf{TISP}(n^{1.1}, n^{0.1})$, then F could be computed in $\mathbf{TISP}(n^{1.1}\text{polylog}(n), n^{0.1}\text{polylog}(n))$.
3. But then one would have $L \in \mathbf{TISP}(n^{1.2}, n^{0.2})$.

The proof of $\mathbf{NTIME}(n) \not\subseteq \mathbf{TISP}(n^{1.2}, n^{0.2})$ is given next.

The Cook-Levin reduction makes use of the configuration circuit.

$$\mathbf{TISP}(n^{12}, n^2) \subseteq \Sigma_2 \mathbf{TIME}(n^8).$$

Suppose L is decided by $\mathbb{M}$ using n^{12} time and n^2 space.

- ▶ Given input x a node of $G_{\mathbb{M},x}$ is of length $O(n^2)$.
- ▶ $x \in L$ iff C_{accept} can be reached from C_{start} in n^{12} steps.
- ▶ There is such a path iff **there exist** n^6 nodes $C_1, \dots, C_{n^6}$, whose total length is $O(n^8)$, such that, **for all** $i \in \{1, \dots, n^6\}$, C_i can be reached from C_{i-1} in $O(n^6)$ -steps.
- ▶ The latter condition can be verified in $O(n^6 \log n)$ -time by resorting to a universal machine.

It is now easy to see that $L \in \Sigma_2 \mathbf{TIME}(n^8)$.

If $\mathbf{NTIME}(n) \subseteq \mathbf{TIME}(n^{1.2})$ then $\Sigma_2\mathbf{TIME}(n^8) \subseteq \mathbf{NTIME}(n^{9.6})$.

Suppose $L \in \Sigma_2\mathbf{TIME}(n^8)$. Then some c, d and $(O(n^8))$ -time TM $\mathbb{M}$ exist such that $x \in L$ iff

$$\exists u \in \{0, 1\}^{c|x|^8}. \forall v \in \{0, 1\}^{d|x|^8}. \mathbb{M}(x, u, v) = 1. \quad (1)$$

Given $\mathbb{M}$ one can design a **linear** time **ND**TM $\mathbb{N}$ that given $x \circ u$ returns 1 iff $\exists v \in \{0, 1\}^{d|x|^8}. \mathbb{M}(x, u, v) = 0$.

- ▶ By assumption there is some $O(n^{1.2})$ -time TM $\mathbb{D}$ such that $\mathbb{D}(x, u) = 1$ iff $\exists v \in \{0, 1\}^{d|x|^8}. \mathbb{M}(x, u, v) = 0$.
- ▶ Consequently $\overline{\mathbb{D}}(x, u) = 1$ iff $\forall v \in \{0, 1\}^{d|x|^8}. \mathbb{M}(x, u, v) = 1$.

It follows that there is an $O(n^{9.6})$ time NDTM $\mathbb{C}$ such that

$$\mathbb{C}(x) = 1 \text{ iff } \exists u \in \{0, 1\}^{c|x|^8}. \overline{\mathbb{D}}(x, u) = 1 \text{ iff (1) holds iff } x \in L,$$

implying that $L \in \mathbf{NTIME}(n^{9.6})$.

$$\begin{array}{ll}
\mathbf{NTIME}(n) & \subseteq \mathbf{TISP}(n^{1.2}, n^{0.2}), \text{ hypothesis} \\
& \Downarrow \\
\mathbf{NTIME}(n^{10}) & \subseteq \mathbf{TISP}(n^{12}, n^2) \\
& \Downarrow \\
\mathbf{NTIME}(n^{10}) & \subseteq \Sigma_2 \mathbf{TIME}(n^8), \text{ alternation introduction} \\
& \Downarrow \\
\mathbf{NTIME}(n^{10}) & \subseteq \mathbf{NTIME}(n^{9.6}), \text{ alternation elimination,} \\
& \text{but} \\
\mathbf{NTIME}(n^{9.6}) & \subsetneq \mathbf{NTIME}(n^{10}), \text{ Hierarchy Theorem.}
\end{array}$$

Proof by Indirect Diagonalization

Suppose we want to prove $\mathbf{NTIME}(n) \not\subseteq \mathbf{TISP}(T(n), S(n))$.

1. Assume $\mathbf{NTIME}(n) \subseteq \mathbf{TISP}(T(n), S(n))$.
2. Derive unlikely inclusions of complexity classes.
 - ▶ Introduce alternation to speed up space bound computation.
 - ▶ Eliminate alternation using hypothesis.
3. Derive a contradiction using a diagonalization argument.

Lance Fortnow proved the first time-space lower bound. A survey on the time-space lower bounds for satisfiability is given by Dieter van Melkebeek.

1. Lance Fortnow. Time-Space Tradeoffs for Satisfiability. *Journal of Computer and System Sciences*, 60:337-353, 2000.
2. Dieter van Melkebeek. A Survey of Lower Bounds for Satisfiability and Related Problems. *Foundations and Trends in Theoretical Computer Science*, 2:197-303, 2007.